

IP Addressing, Subnetting & Routing

IPv4 Addresses

An IPv4 address is 32 bits, written as four octets: 192.168.1.100 .

Component	Meaning
Network bits	Identify the network
Host bits	Identify the machine on that network
Subnet mask	Divides the two (e.g., /24 = 255.255.255.0)

CIDR Notation

- 10.0.0.0/8 → 16.7M addresses (large)
- 172.16.0.0/12 → ~1M addresses
- 192.168.0.0/16 → 65,536 addresses (typical home)
- /24 → 256 addresses (common LAN)

The /N means N bits are "network". The rest are "host".

Subnetting Math Rules

With /N mask: $2^{(32-N)}$ addresses per subnet. The first is the network, the last is broadcast. Usable hosts = $2^{(32-N)} - 2$.

Example: 192.168.1.0/24

- Network: 192.168.1.0
- Broadcast: 192.168.1.255
- Usable: 192.168.1.1 – 192.168.1.254

Security use: subnetting/Segmentation is a core defense – you use CIDR to carve the network into zones (IoT /16, staff /24, servers /27). The subnet mask is literally your compartmentalization.

Public vs Private & NAT

Private ranges (RFC 1918) are not routable on the internet:

10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

NAT (Network Address Translation) lets many private hosts share one public IP. The router rewrites the source IP on outbound packets and remembers the mapping (the "connection table") to route replies back.

NAT provides a form of *obscurity* but not real security – and attackers behind NAT still get attacked. Also: NAT tables are a DoS target (entries can be exhausted).

How Routing Works

Each router:

- Receives a packet
- Looks at destination IP
- Consults its **routing table** (destination → interface/next-hop)
- Forwards to the next hop
- Decrements TTL (prevents infinite loops)

Routing tables can be attacked: route hijacking (BGP), on-path attacks, malicious default routes. A compromised router = everything routed through it is compromised.

IPv6 Essentials

IPv6 uses 128-bit addresses (8 groups of 4 hex digits). Notable:

- Huge address space (no NAT needed typically)
- SLAAC for autoconfiguration
- Stateless address autoconfiguration → devices auto-appear on networks

IPv6 is often forgotten in security policies. Attackers frequently use IPv6 tunnels to bypass IPv4-only firewalls. If you block `::` but allow `0.0.0.0/0`, attackers can tunnel out.

[← TCP, UDP & Ports in Depth](#)

[DNS in Depth & DNS Attacks →](#)